

AeroGrid Turbine Telemetry Engineering Diagnostic Report

Addressed to: Chief Technology Officer, AeroGrid

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Findings:

The analysis of the telemetry dataset of 5000 readings across 10 turbines, uncovered two malfunctioning turbines, requiring urgent care:

Turbine T-04 — Average temperature of **90.58°C**, exceeding the 85.0°C safety threshold, indicating a sustained overheating condition consistent with a probable cooling system fault or bearing failure.

Turbine T-07 — Maximum vibration spike of **25.0 mm/s**, far exceeding the 15.0 mm/s threshold. The level of extreme oscillation suggests a structural imbalance or probable blade damage, proving to be a significant threat if left unaddressed.

Turbines: T-01, T-02, T-03, T-05, T-06, T-08, T-09, T-10 are all operating within safe parameters.

Architecture proposal and justification:

This proposal covers a potential replacement to an existing single legacy server, proving to being inefficient, the improved system includes a cloud-based architecture to combat rising tensions of the old system.

AWS Kinesis acts as a message queue, buffering data traffic from all turbines simultaneously, holding data safely and navigating data to the AWS Lambda for processing. If downstream processing slows, this eliminates crash risk.

AWS Lambda process and applies anomaly detection rules in real time as data flows through, triggering instant engineer alerts when certain parameters are breached.

InfluxDB stores recent data optimised for fast dashboard queries. **AWS S3** archives older data cheaply as a long-term data lake (cold data). This separation ensures the system never loses data under peak load.

Cost optimisation

Data storage can be optimised through implementation of S3 intelligent-Tiering policy. This policy enables the renewal of data into cheaper storage tiers if it has not been accessed for 30 days. This in turn reduces cold storage costs by up to 40%, all whilst being autonomous – critical for a pipeline of this magnitude.

